

ADDRESSING MODES

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WHY ADDRESSING MODES?

To address memory in different ways as and where needed!

How many modes?

In 8086/88, there are seven modes to address the memory;

1. Register Addressing

MOV AX, BX	(legal)
ADD AL, BL	(")
MOV DS, AX	(")
must be { MOV AL, BX	(Illegal)
of same size { MOV DX, CL	(")

2. Immediate Addressing

MOV AL, 20H	(Legal)
ADD BL, 7FH	(")
MOV AX, 2000H	(")
Constant value cannot be moved directly into Segment Registers [MOV CS, 470FH]	(Illegal)

alternate
↓

MOV AX, 470FH
MOV CS, AX

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3. Direct Addressing

MOV [0200], AL

Move a byte from 'AL' into memory location address ~~0200H~~ 0200H.

[] → required to make the value (e.g., 200) address location

4. Indirect Addressing

MOV BX, 0200H

MOV [BX], AL

Same ~~result~~ result as '3', However the address is in 'BX' register

Registers that can be used for indirect addressing:

Register	Default Segment
BX	DS
SI	DS, ES
DI	DS, ES
IP	CS
BP	SS
SP	SS

NOTE:

This addressing mode is applied for implementation of one-dimensional array

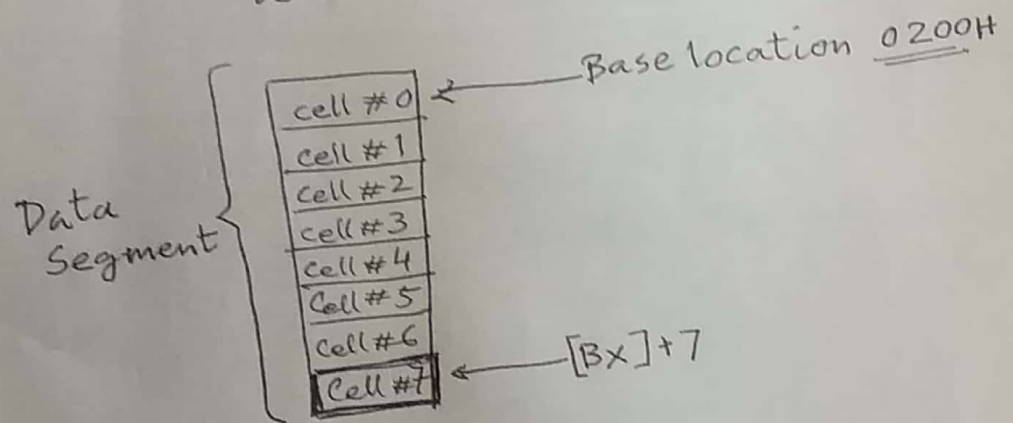
5. Based Relative Addressing:

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Using 'Bx' or 'BP' as base pointers to access a particular address location in 'DS' or 'SS' relatively.

It is used to access a specific byte or word in the memory.

For example, if cell #7 is to be accessed, its location will be defined by the base + 7 relative displacement



`MOV BX, 0200H`

`MOV AL, [BX] + 7; AL = DS:[0200 + 7]`

for stack segment, 'BP' will be used as default pointer, e.g.;

`MOV BP, 0200H`

`MOV AL, [BP] + 7; AL = SS:[0200 + 7]`

6. Indexed Relative Addressing

Same concept as "Based Relative Addressing mode". The difference is the use of Index registers 'SI' and 'DI' as pointer instead of 'BX' and 'BP'.

'SI' is default index register for DS.
'DI' is default index register for ES.

7. Based Indexed Addressing

It is used to access a specific location (or cell) in two dimensional array.

For example, consider the following dataset

	Col0	Col1	Col2	Col3
	Courses			
Roll#	English	Physics	Chemistry	Maths
Row0 001				
Row1 002				
Row2 003				
Row3 004				

[Row1][Col1]

Here; BX = Row1
SI = Col1

We can use DI = Col1.

MOV BX, Row1
MOV SI, Col1
MOV AL, [BX][SI] ; BX=Row1
SI=Col1

for stack
use BP
instead of BX